Signals and Systems

Computer Assignment 5

Department of Electrical Engineering

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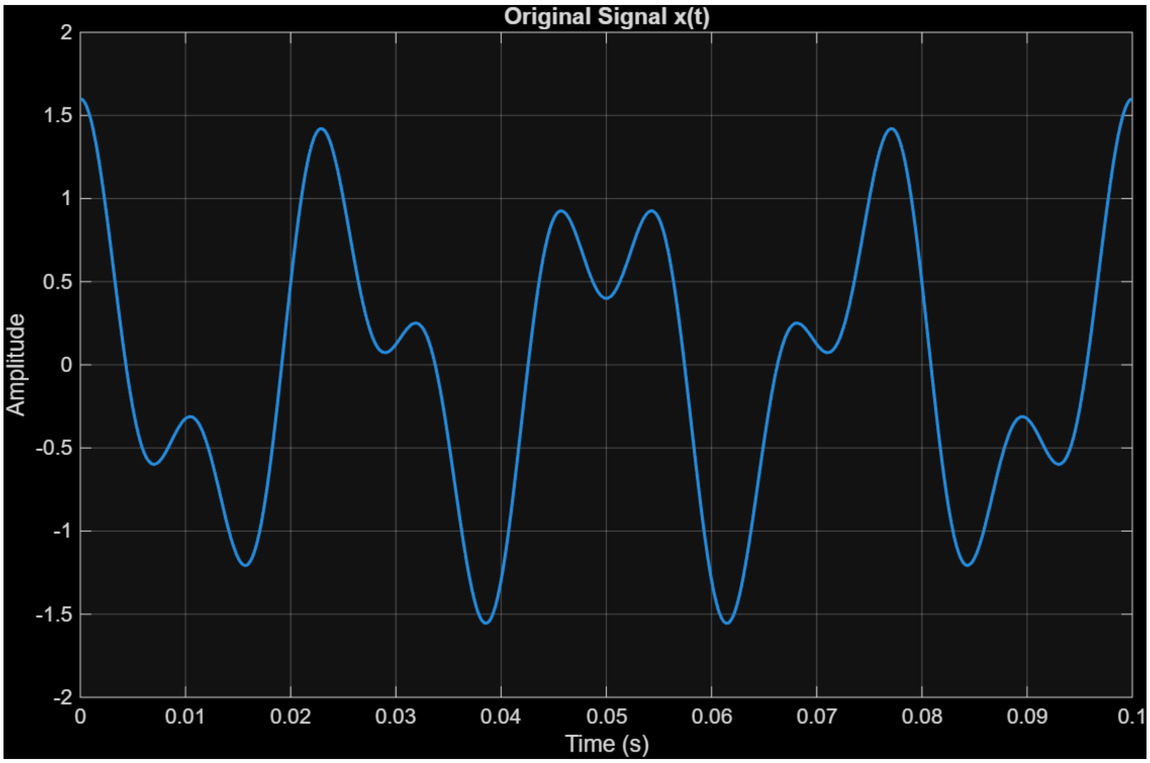
Contents

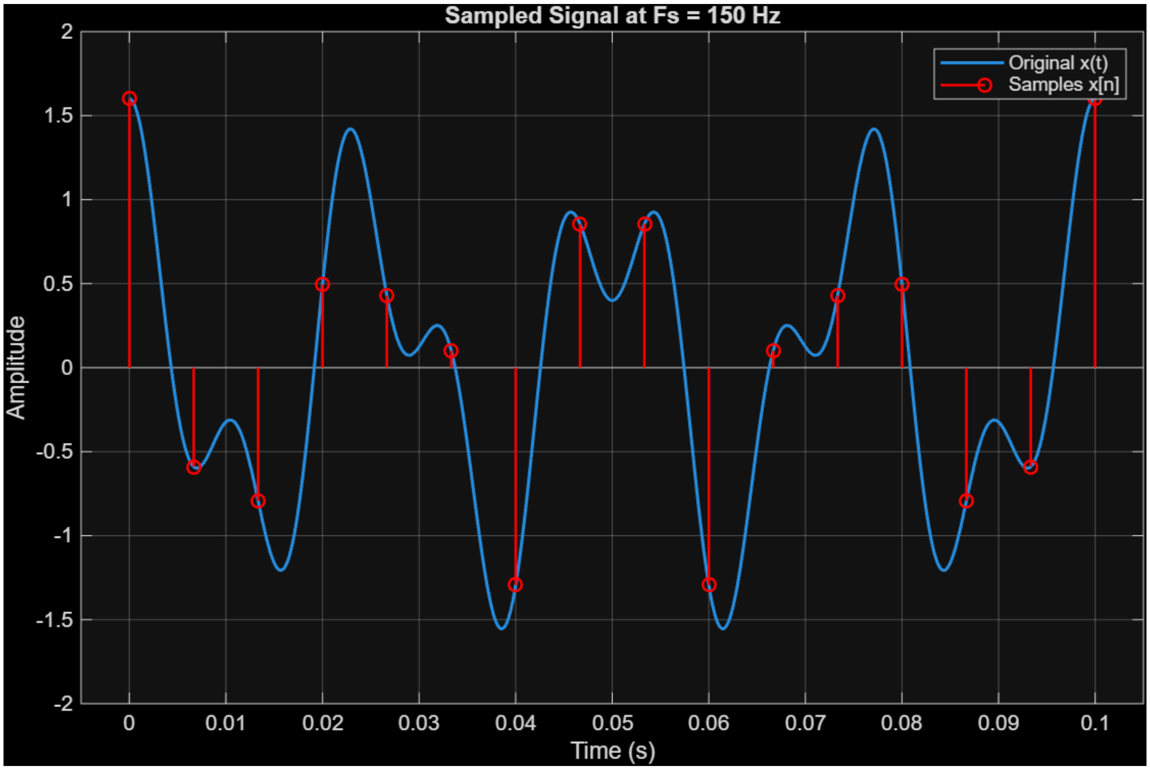
1. Ideal Sampling and Sinc Reconstruction

The continuous-time signal is given as $x(t) = \cos(2\pi 40t) + 0.6\cos(2\pi 90t)$.

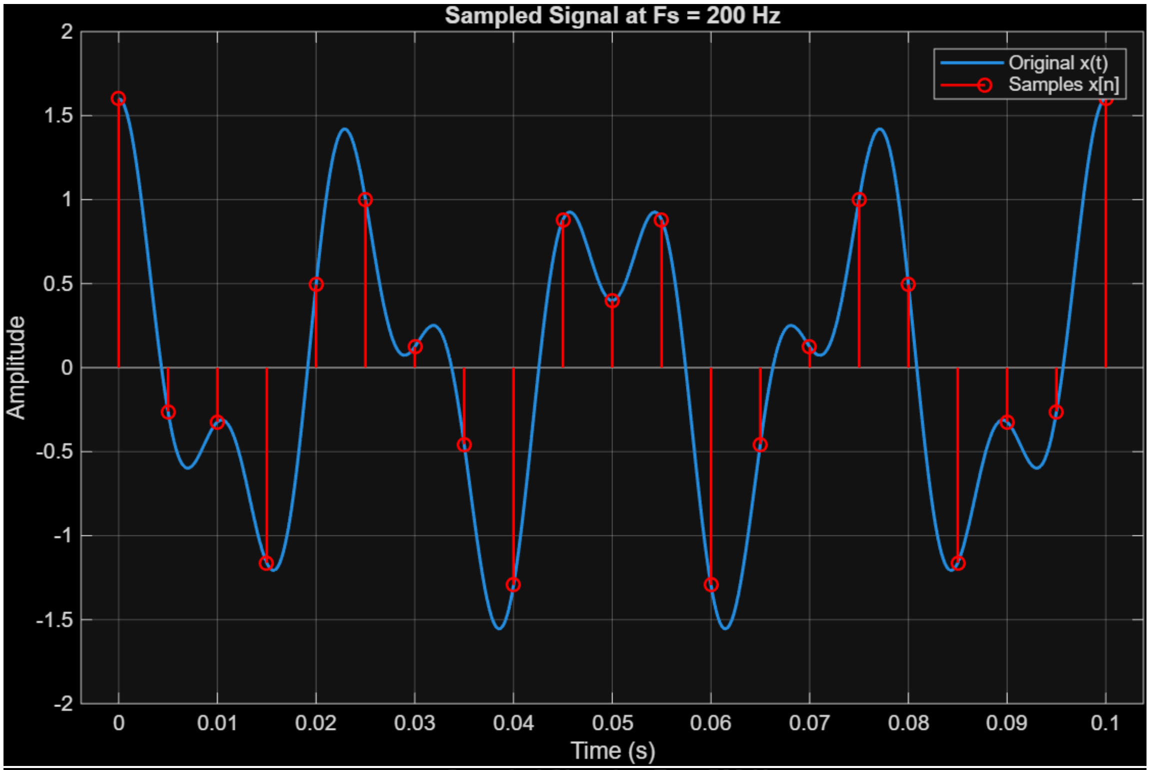
- **(a) Nyquist Rate:** The signal consists of two frequency components: 40 Hz and 90 Hz. The highest frequency present in the signal is $f_{max} = 90$ Hz.According to the Nyquist-Shannon sampling theorem, the theoretical Nyquist sampling rate is $2 \times f_{max} = 180$ Hz.

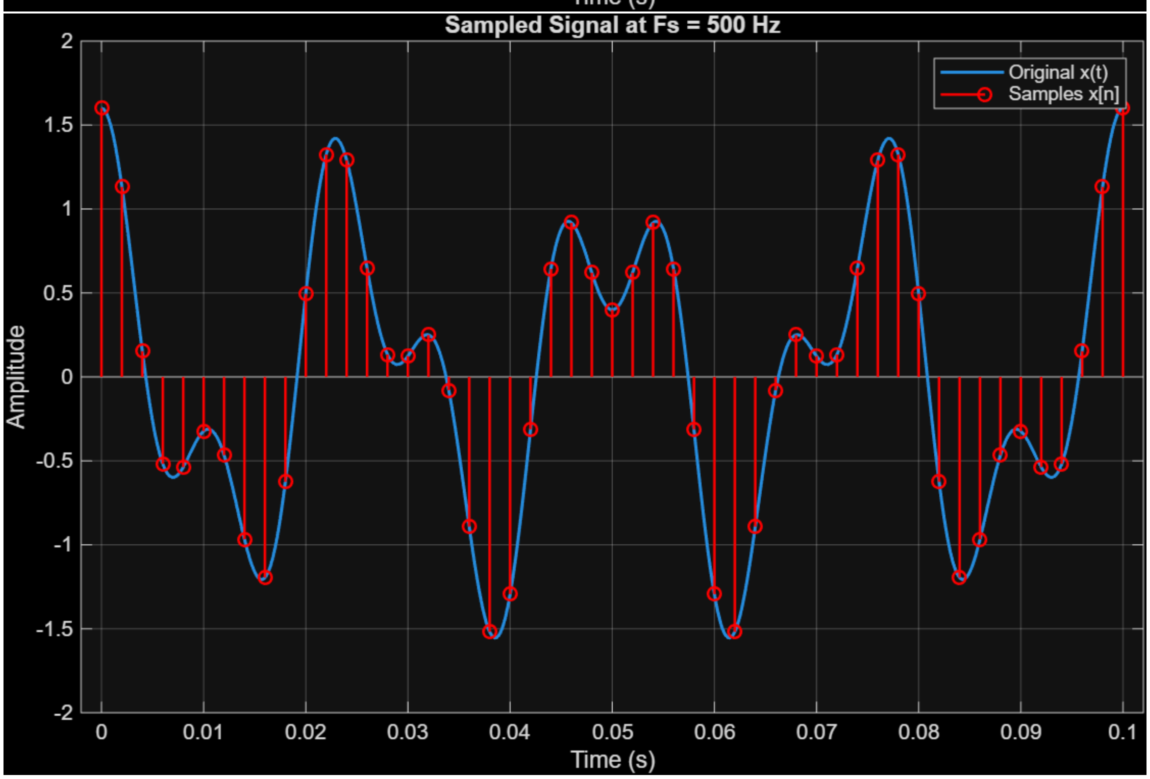


- (b)

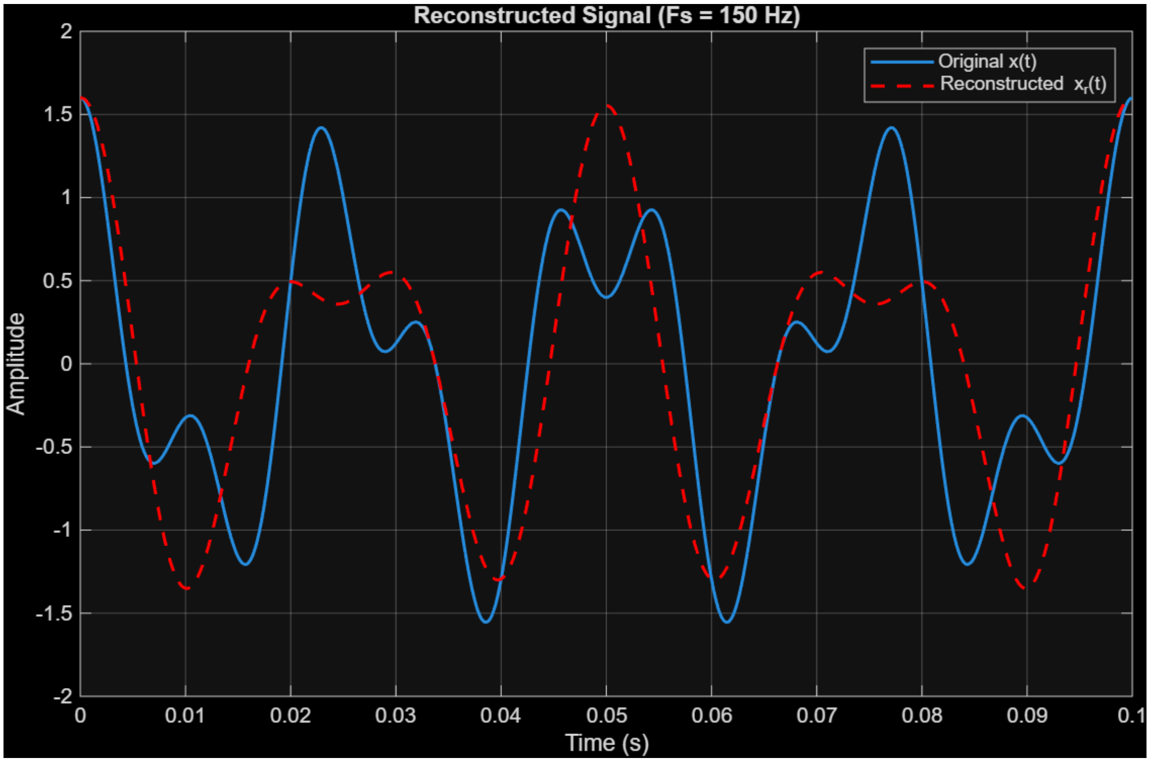


Sampled Signal at Fs = 150 Hz
Original x(t)
Samples x[n]
Amplitude
Time (s)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1

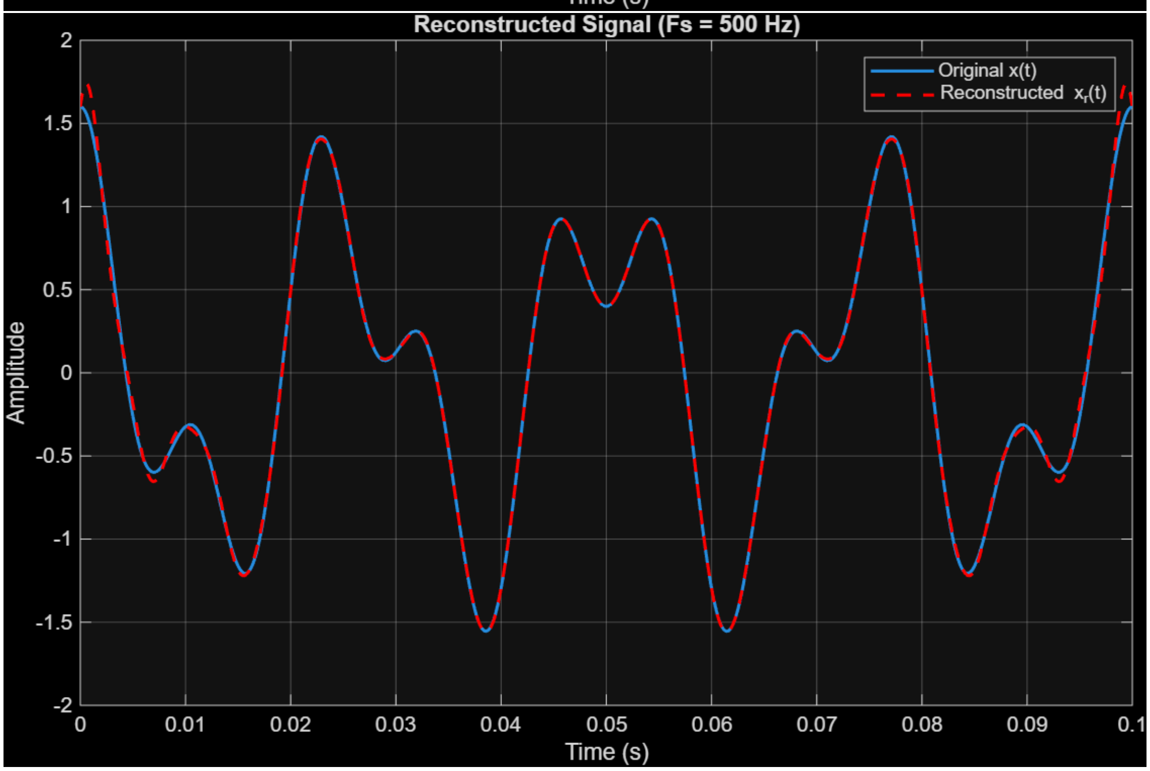

Sampled Signal at Fs = 200 Hz
Original x(t)
Samples x[n]
Amplitude
Time (s)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1


Sampled Signal at Fs = 500 Hz
Original x(t)
Samples x[n]
Amplitude
Time (s)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1

- **(c)**


Reconstructed Signal (Fs = 150 Hz)
Original x(t)
Reconstructed $x_r(t)$
Amplitude
Time (s)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1


Reconstructed Signal (Fs = 200 Hz)
Original x(t)
Reconstructed $x_r(t)$
Amplitude
Time (s)


Reconstructed Signal (Fs = 500 Hz)
Original x(t)
Reconstructed $x_r(t)$
Amplitude
Time (s)

- **(d) Reconstruction Accuracy:** We sampled the signal at $F_s = 150$ Hz, 200 Hz, and 500 Hz. For accurate reconstruction using ideal sinc interpolation, the sampling frequency must satisfy $F_s > 180$ Hz.
  - At $F_s = 150$ Hz, the sampling rate is below the Nyquist rate. This results in aliasing, making accurate reconstruction impossible (high MSE).
  - At $F_s = 200$ Hz and $F_s = 500$ Hz, the condition $F_s > 180$ Hz is met. Therefore, the signal is reconstructed accurately with an error close to zero.

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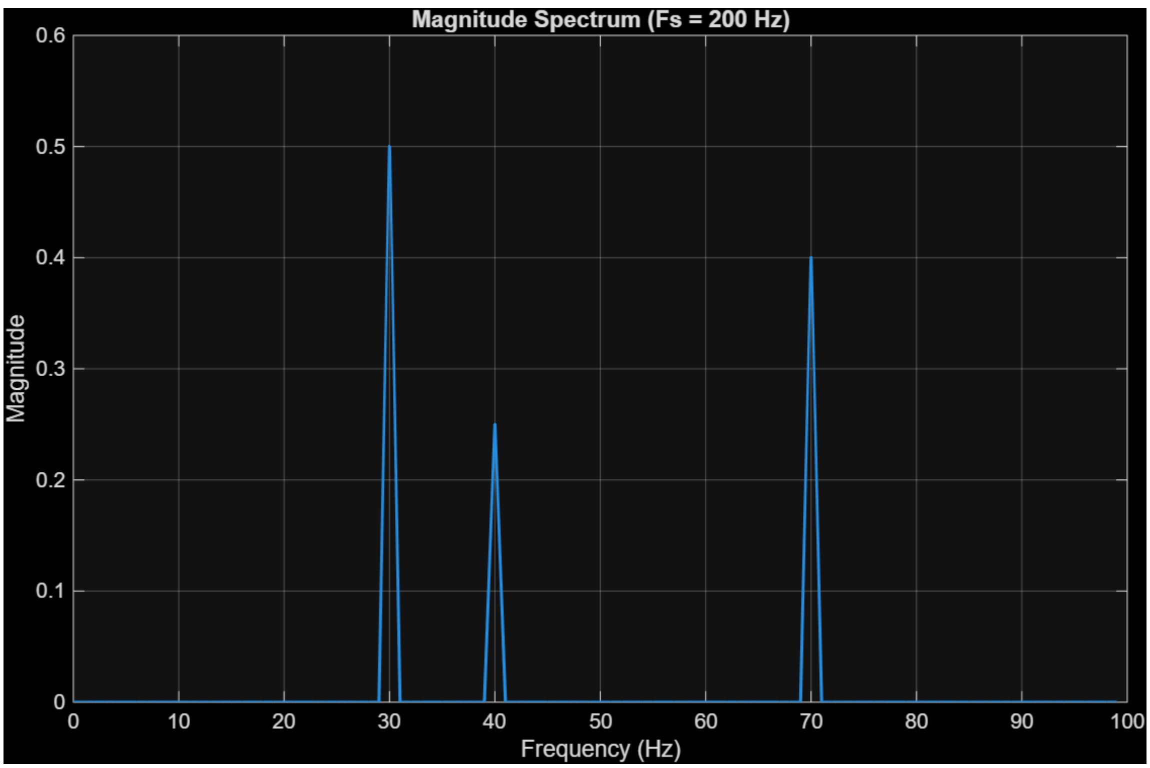
Mean Squared Errors for Fs = 150, 200, 500 Hz:
    0.3759    0.0001    0.0014
```

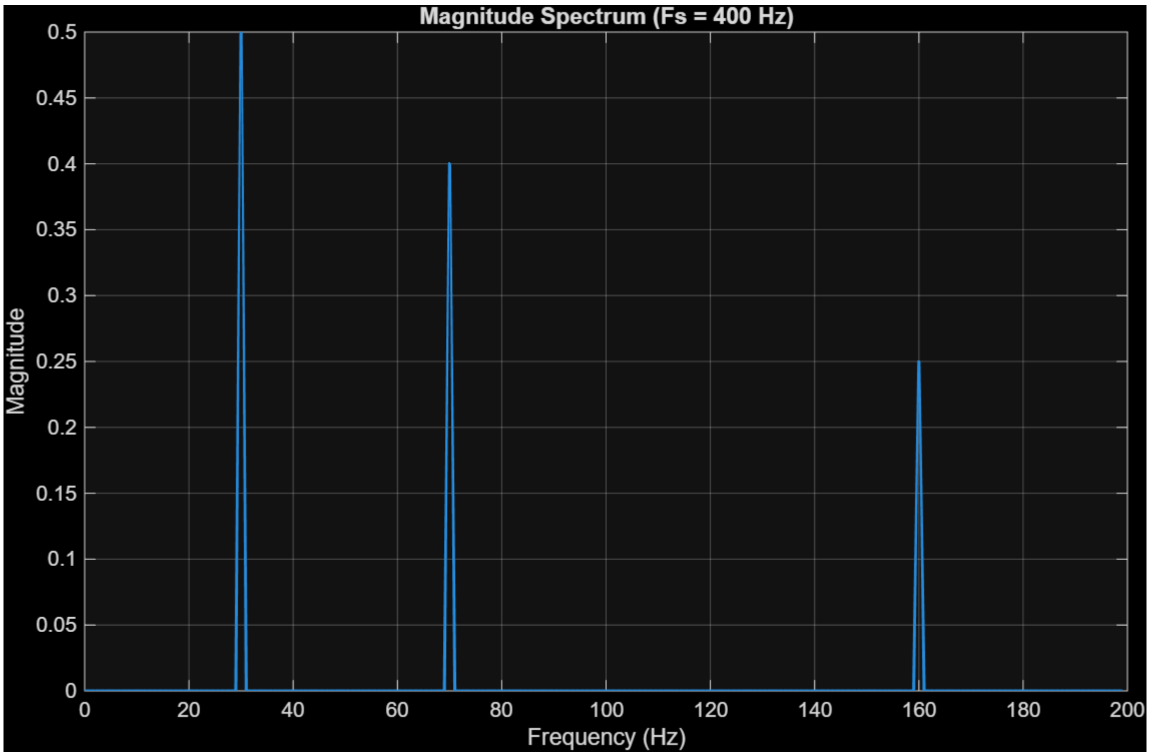
2. Aliasing in the Frequency Domain

The signal is $x(t) = \cos(2\pi 30t) + 0.8\cos(2\pi 70t) + 0.5\cos(2\pi 160t)$.

- **(a) Nyquist Rate:** The highest frequency component is 160 Hz. The Nyquist rate is $2 \times 160 = 320$ Hz.
- **(b) Sampling at 200 Hz:** The folding frequency is $F_s/2 = 100$ Hz.
  - The 30 Hz and 70 Hz components are below 100 Hz, so they appear correctly at 30 Hz and 70 Hz.
  - The 160 Hz component exceeds the folding frequency and causes aliasing. Its apparent frequency becomes $|160 - 200| = 40$ Hz.
- **(c)**



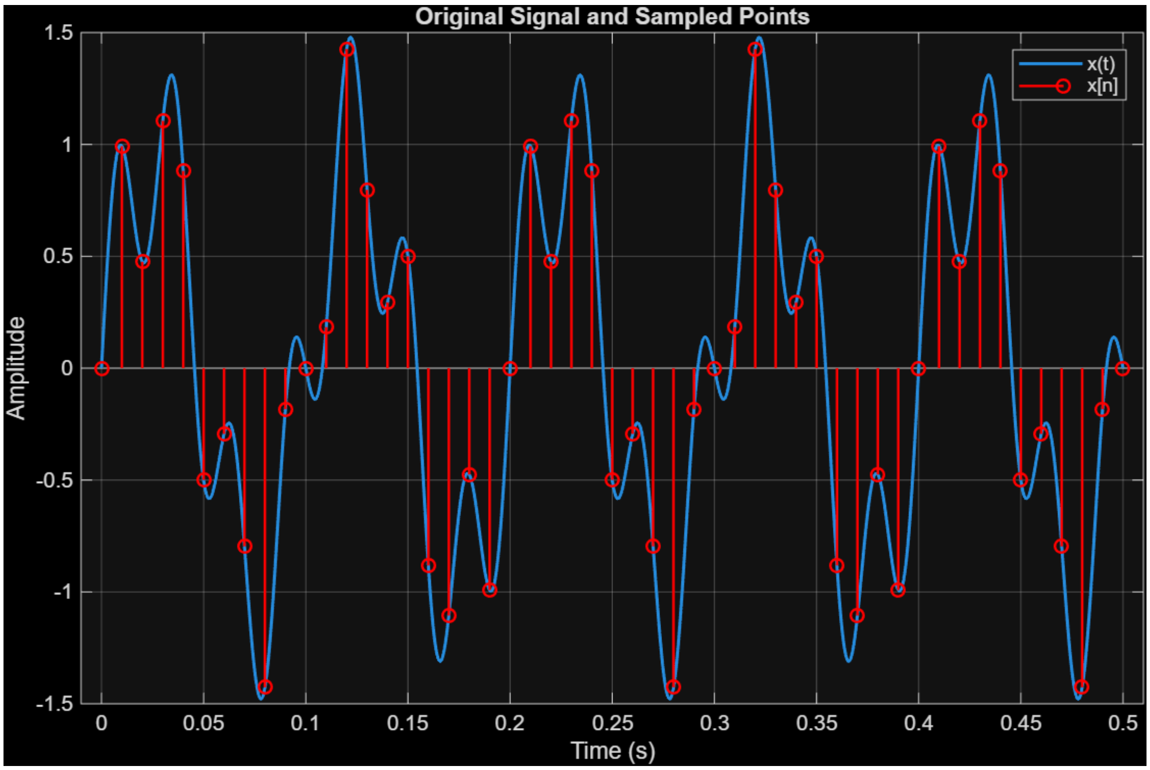
- **(d) Sampling at 400 Hz:** Since $F_s = 400$ Hz is greater than the Nyquist rate of 320 Hz, no aliasing occurs. The spectrum correctly shows peaks at 30, 70, and 160 Hz.



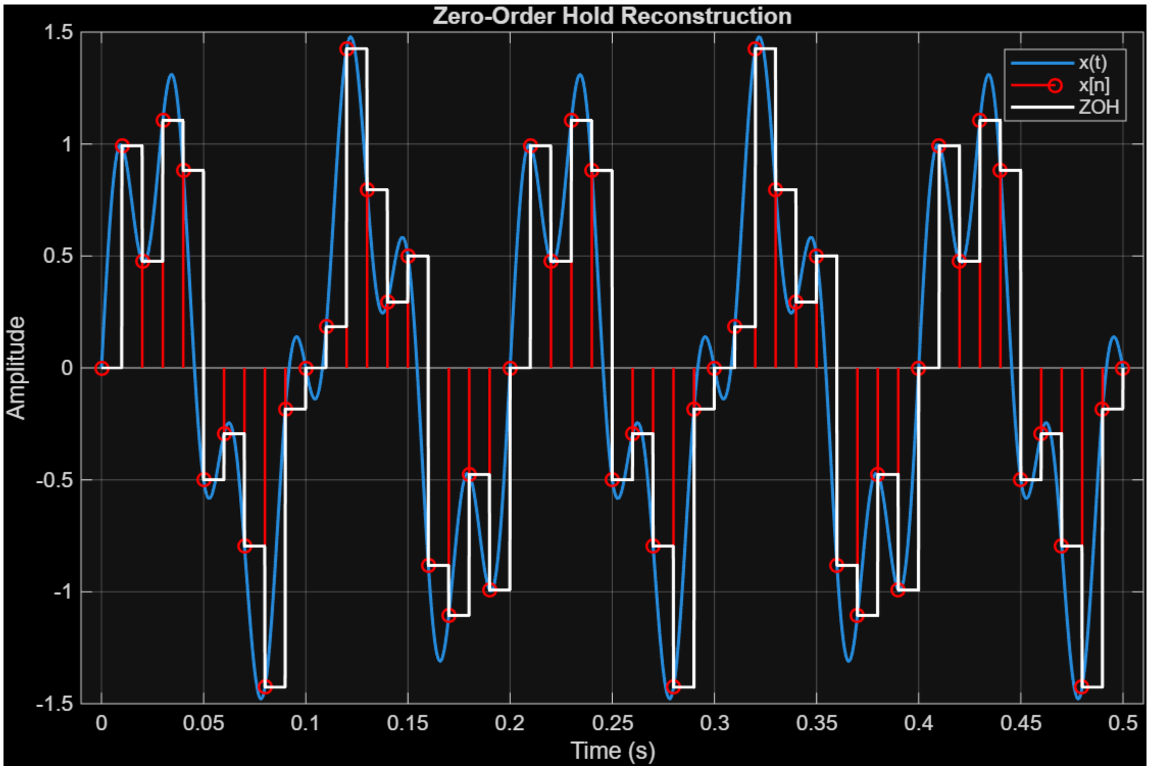
3. Practical Sampling with Zero-Order Hold

The signal is $x(t) = \sin(2\pi 10t) + 0.5\sin(2\pi 35t)$, sampled at $F_s = 100$ Hz.

- **(a)**

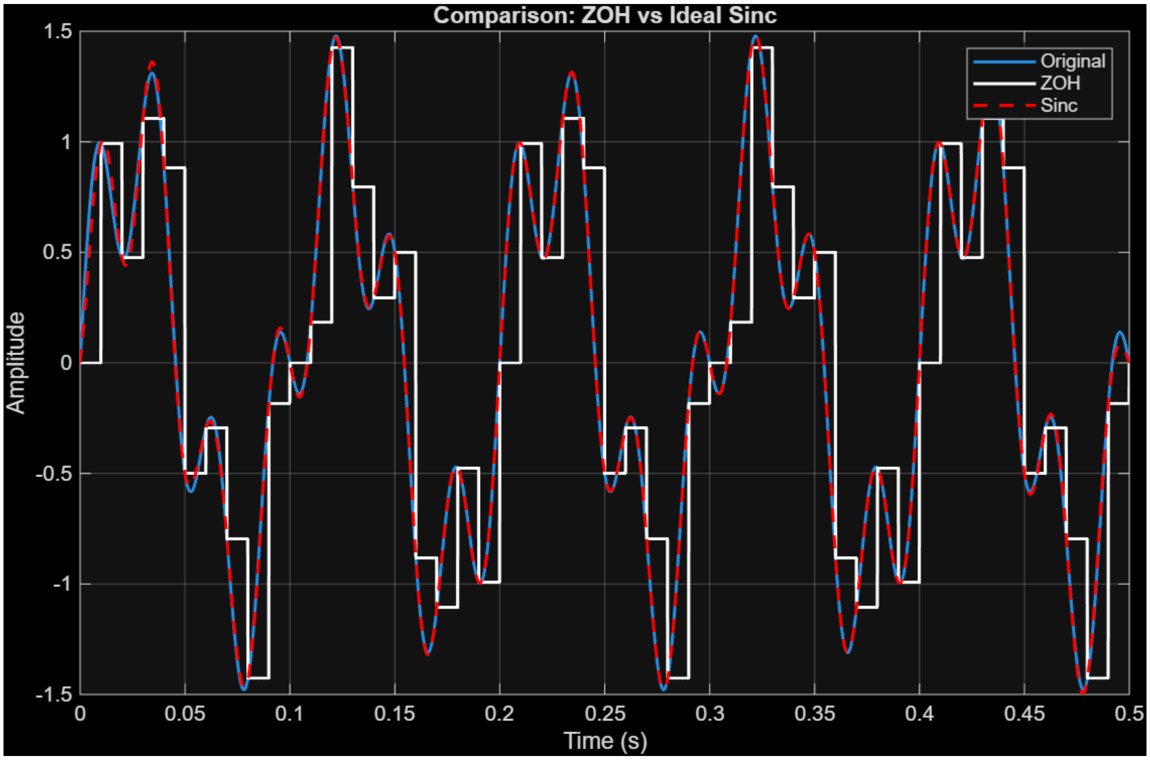


- **(b)**



Zero-Order Hold Reconstruction
x(t)
x[n]
ZOH
Amplitude
Time (s)
1.5
1
0.5
0
-0.5
-1
-1.5
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

- **(c)**


Comparison: ZOH vs Ideal Sinc
Original
ZOH
Sinc
Amplitude
Time (s)
1.5
1
0.5
0
-0.5
-1
-1.5
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

- **(d)**

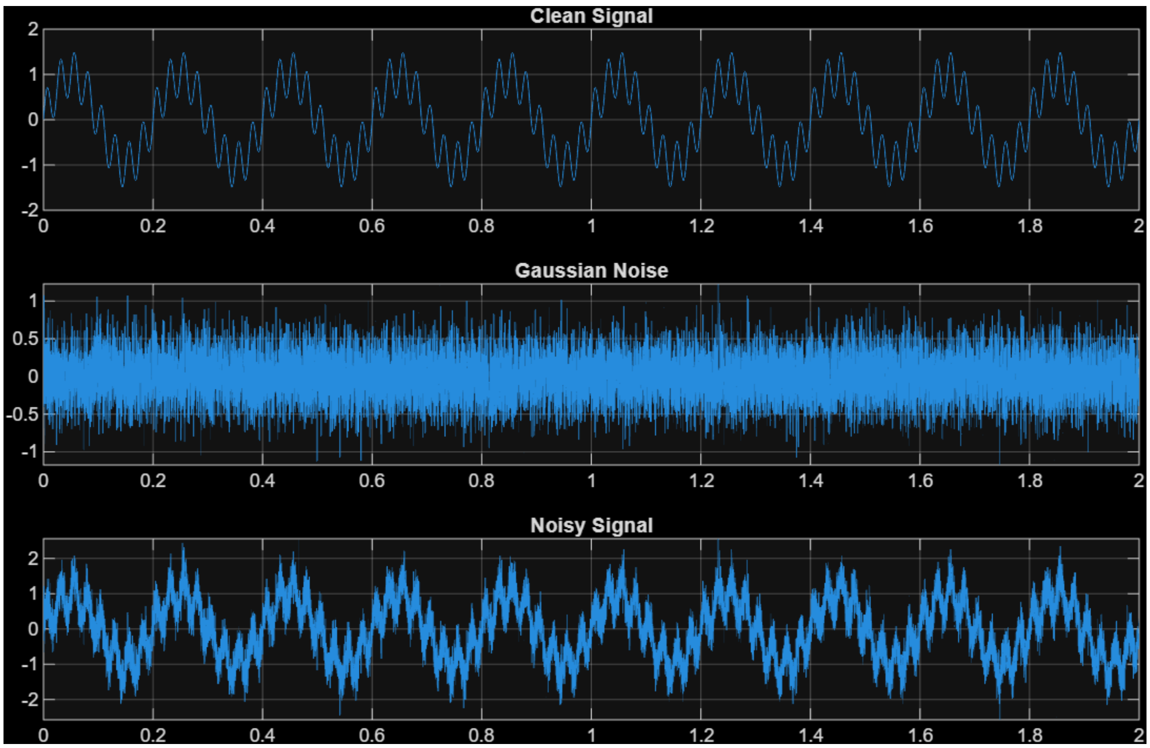
```
Reconstruction Error (MSE):
ZOH Error: 0.22293
Sinc Error: 0.00063245
```

ZOH vs. Ideal Sinc: Zero-Order Hold (ZOH) reconstruction is easier to implement in practical hardware because it only requires a simple hold circuit (like a capacitor) to maintain the voltage level until the next sample. However, it is less accurate than ideal sinc interpolation. ZOH creates a staircase-like continuous signal, which introduces unwanted high-frequency harmonics. Ideal sinc interpolation, on the other hand, acts as a perfect low-pass filter, resulting in flawless reconstruction with zero error, assuming the Nyquist criterion is met.

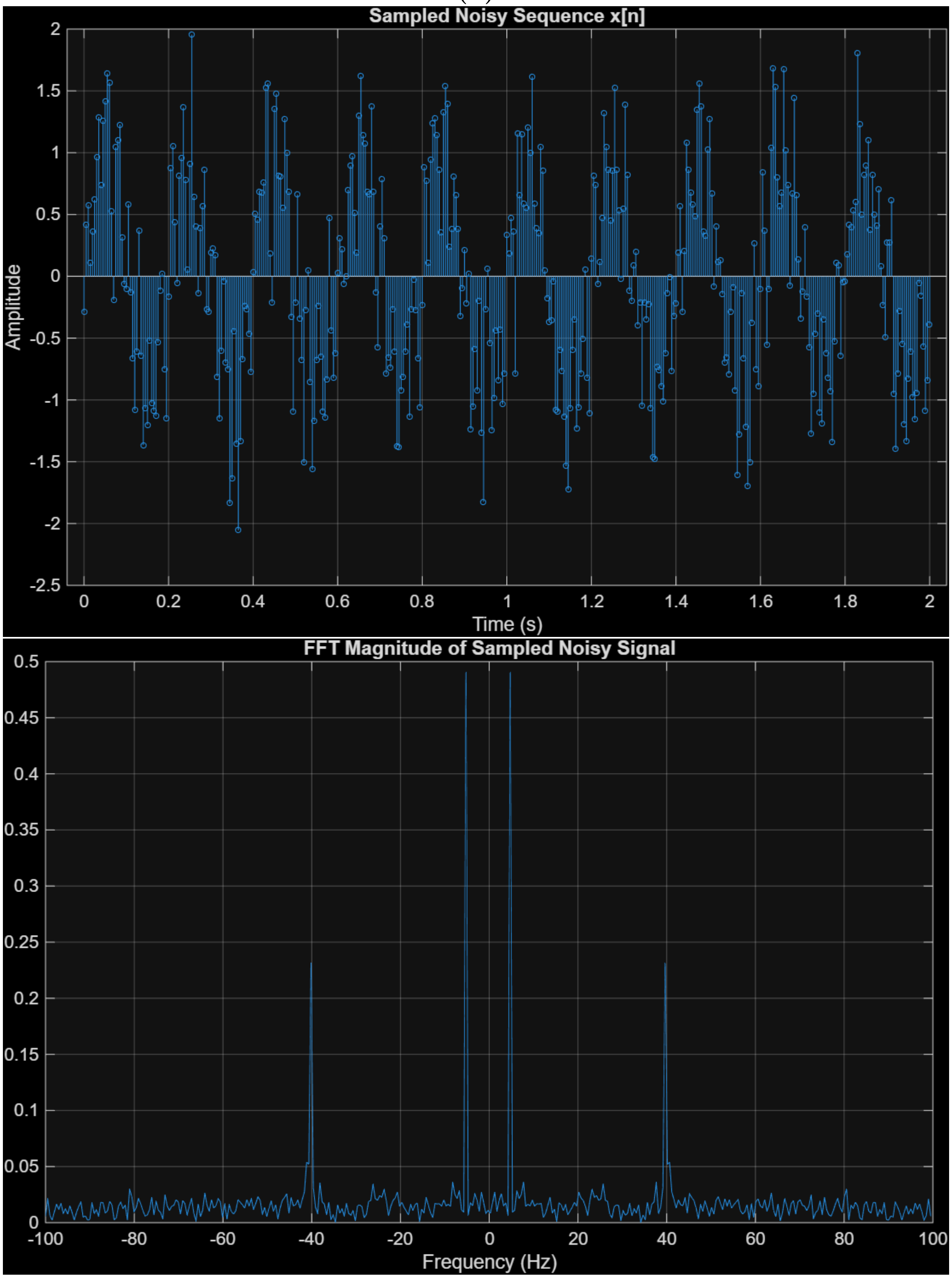
4. Sampling, Digital Processing, and Reconstruction

The signal $x(t) = \sin(2\pi 5t) + 0.5\sin(2\pi 40t) + 0.3w(t)$ is contaminated with Gaussian white noise $w(t)$.

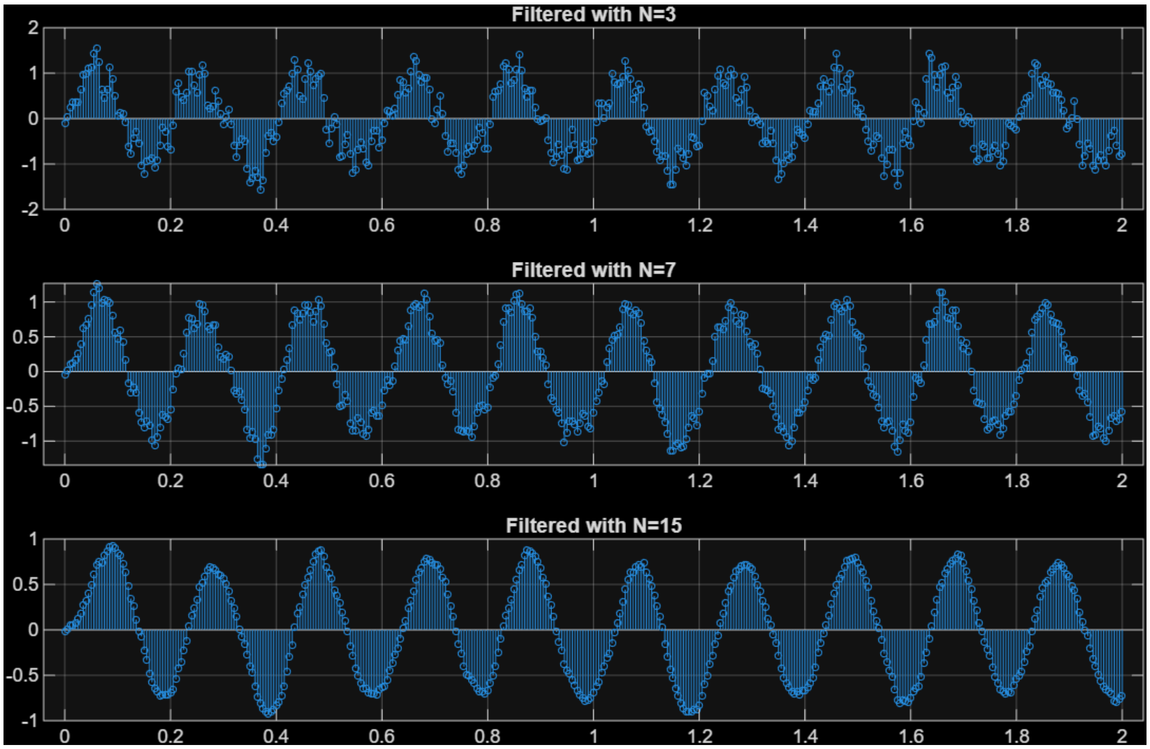
- **(a)**



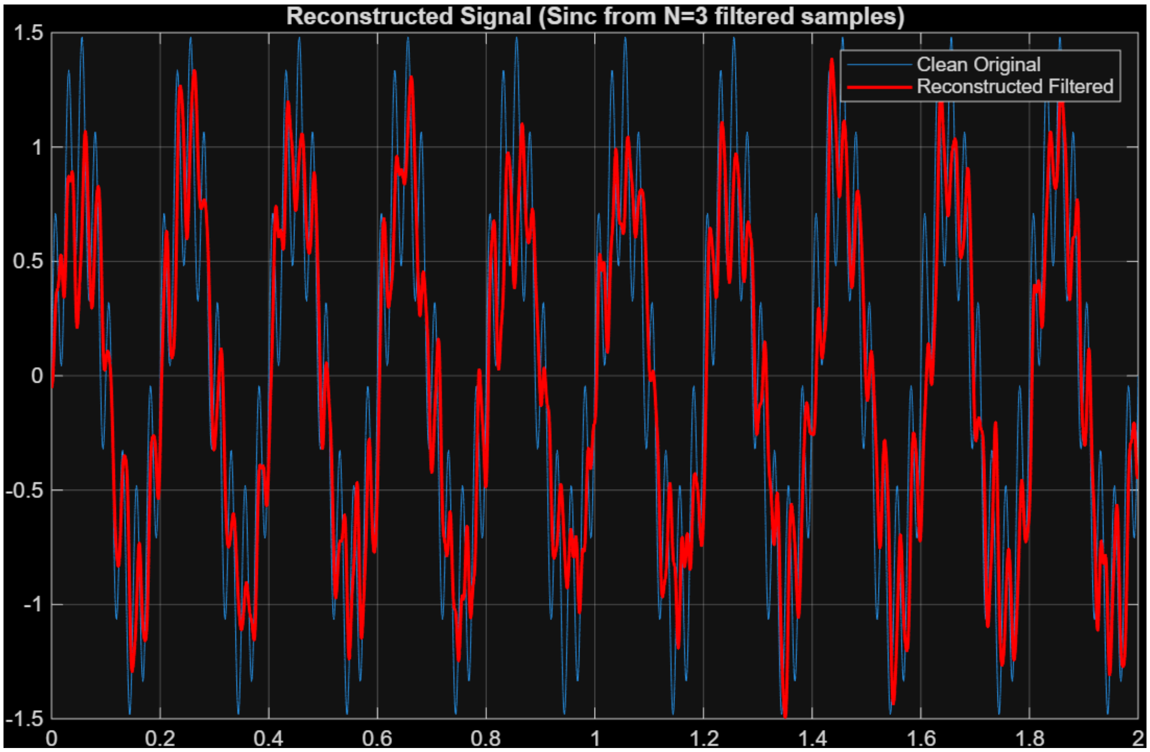
- **(b)**


Sampled Noisy Sequence x[n]
Amplitude
Time (s)
FFT Magnitude of Sampled Noisy Signal
Frequency (Hz)

- **(c) Moving Average Filter:** The moving average filter $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$ acts as a digital low-pass filter. As the filter length N increases (from 3 to 7 to 15), the filtering effect becomes stronger. This increases the smoothness of the signal by heavily attenuating the high-frequency noise. However, a larger N also attenuates the 40 Hz high-frequency component of the clean signal.



- **(d) Reconstruction of Filtered Signal:** Performing digital filtering after sampling modifies the discrete sequence. When this filtered sequence is reconstructed back into a continuous-time signal using sinc interpolation, the resulting signal behaves exactly as if the original continuous-time signal had been passed through an equivalent analog continuous-time low-pass filter.

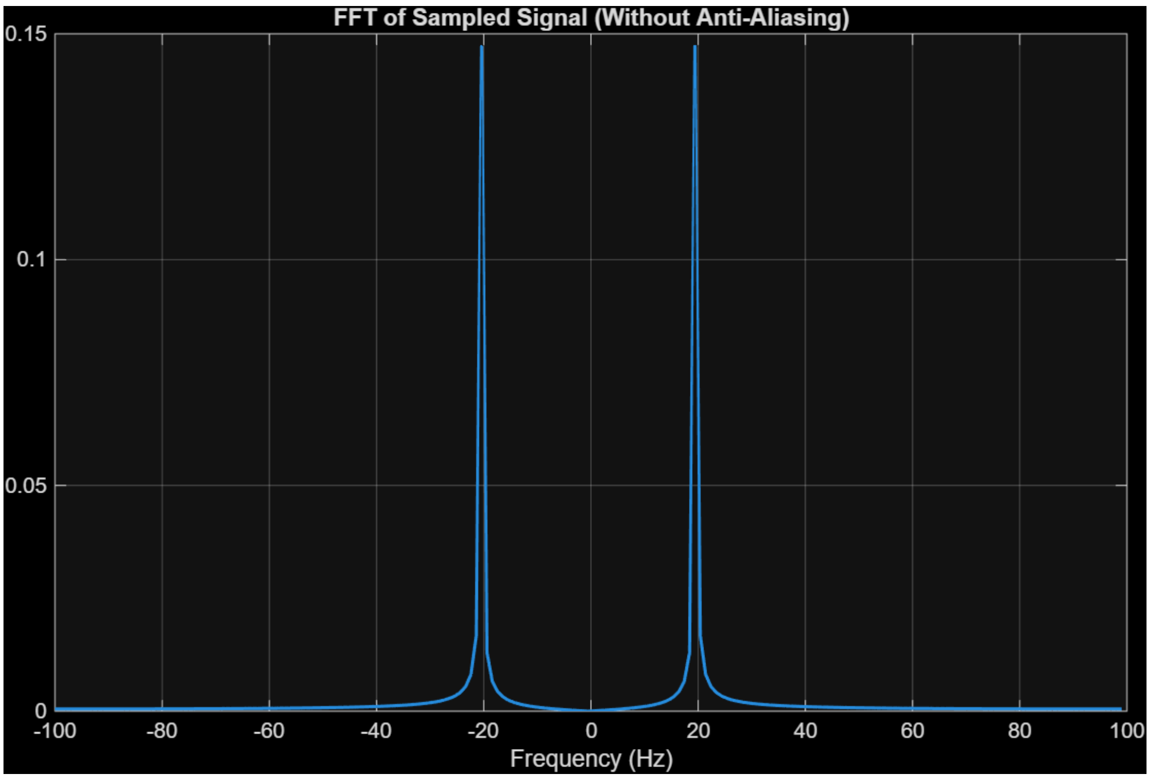


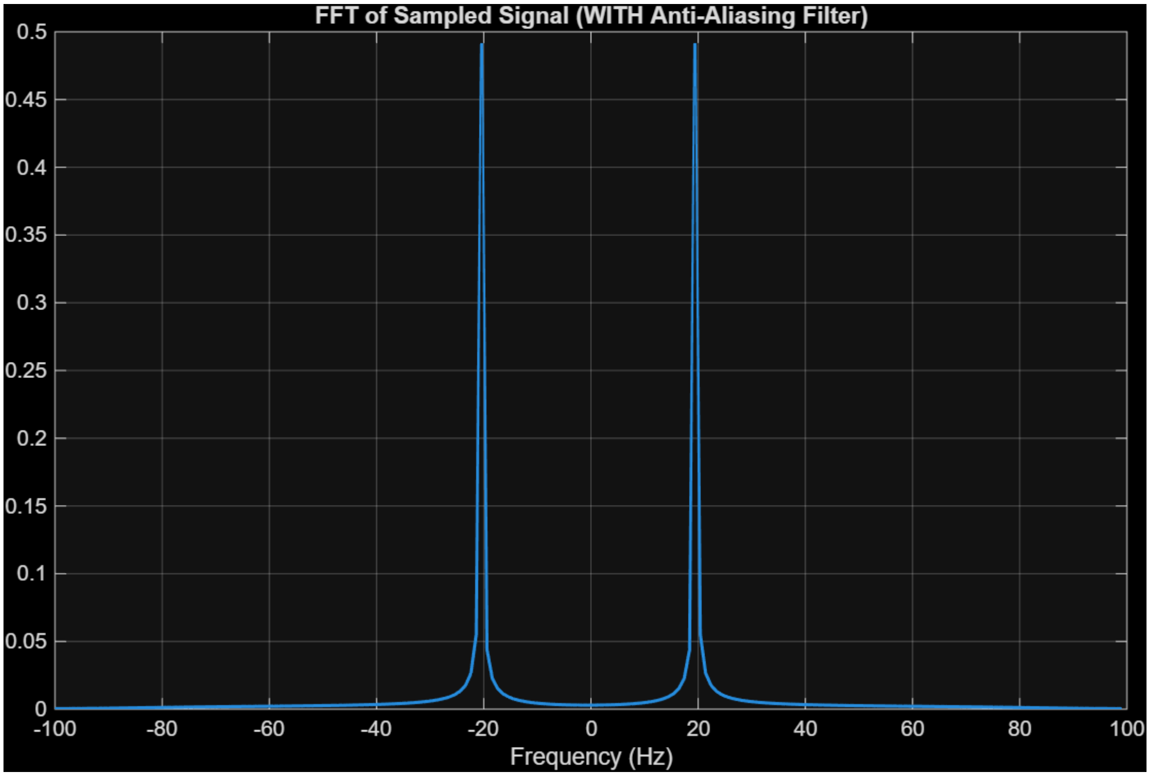
5. Application: Anti-Aliasing Before Sampling

The signal $x(t) = \sin(2\pi 20t) + 0.7\sin(2\pi 180t)$ is sampled at $F_s = 200$ Hz.

- **(a) Theoretical Aliasing:** The folding frequency is $F_s/2 = 100$ Hz. The 180 Hz component is greater than 100 Hz, meaning it violates the Nyquist criterion. It will alias to $|180 - 200| = 20$ Hz. This means the 180 Hz component will perfectly overlap with the 20 Hz component, corrupting the original data.

- **(b),(c),(d)**





- **(e) Necessity of Anti-Aliasing Filters:** Anti-aliasing filters are strictly necessary in real sampling systems. Even if the desired signal's frequency is known, real-world environments contain out-of-band high-frequency noise and interference. If an analog low-pass filter is not applied before sampling, these unwanted frequencies will fold into the baseband due to aliasing. Once aliasing occurs in the discrete domain, it is mathematically impossible to distinguish the aliased noise from the true baseband signal.